

rubai

AI Goal Coach — Full System Specification

A complete documentation snapshot of every layer, package, route, table, file, and tech choice currently present in the rubai monorepo.

Document metadata

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Repository layout: pnpm workspace · TypeScript ~5.9.2

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1. Product

- **Name (wordmark):** rubai (always lowercase). Internal identifiers Atlas* / RubAI are historical.
- **Type:** Mobile-first AI goal-execution coach. User picks a goal (template or custom), AI runs intake !' builds a multi-phase roadmap !' generates a daily plan !' coaches them every day with adaptive evolution.
- **Tiers:** Free (1 active goal), Pro (5), Premium (25). Currently UI-preview only — billing is wired structurally but no charges.
- **Target platforms:** iOS + Android via Expo Go for dev; production-ready EAS build path. Web is dev preview only.

2. Repository Layout (pnpm monorepo)

```
artifacts/
  api-server/      # Express API (mounted at /api)
  mobile/          # Expo React Native app
  mockup-sandbox/  # Vite component-preview sandbox
lib/
  api-spec/        # OpenAPI 3.1 + Orval codegen
  api-zod/         # Generated Zod schemas (server)
  api-client-react/ # Generated React Query hooks (mobile)
  db/              # Drizzle ORM schema + push CLI
  integrations-openai-ai-server/ # OpenAI server wrapper
  integrations-openai-ai-react/  # OpenAI client helpers (audio)
scripts/          # Repo-level utility scripts
pnpm-workspace.yaml # Workspace + catalog + 1-day quarantine
tsconfig.base.json # Shared strict TS defaults
tsconfig.json      # Solution file for libs only
```

3. Tech Stack & Versions

- **Language / build:** TypeScript ~5.9.2, esbuild (server bundle), Metro (mobile), Vite 7 (mockup sandbox).
- **Runtime:** Node 20 (Replit container, NixOS).
- **Mobile:** Expo SDK 54.0.27 · React Native 0.81.5 · React 19.1.0 · expo-router 6 · React Compiler enabled · newArchEnabled: true.
- **Server:** Express 5 · pino 9 (structured logs, no console.log) · cookie-parser · cors · multer (audio uploads) · http-proxy-middleware · express-rate-limit 8.
- **Auth:** Clerk (@clerk/expo 3.2 mobile, @clerk/express 2.1 server). Email+password and SSO. Custom branded screens.
- **DB:** PostgreSQL via pg 8 + Drizzle ORM 0.45 + drizzle-kit (push migrations). drizzle-zod for schema !' Zod.
- **Validation / contracts:** OpenAPI 3.1 + Orval 8.5 codegen !' Zod (server) + React Query hooks (mobile).
- **State (mobile):** React Query 5.90 + AsyncStorage (cache snapshot) + a single AtlasProvider context.
- **Icons / fonts:** @expo/vector-icons (Feather, Ionicons, MaterialIcons), Inter via @expo-google-fonts/inter. .ttf files copied into assets/fonts/ and loaded by direct require to defeat Android Expo Go bundling bug.
- **Animations / UI:** react-native-reanimated 4.1 · gesture-handler · keyboard-controller · expo-linear-gradient · expo-blur · expo-glass-effect · expo-haptics · expo-symbols (SF Symbols on iOS).
- **Media:** expo-image-picker 17, expo-image 3, expo-audio (mic), expo-speech (TTS).
- **Push:** expo-notifications + expo-device (client); expo-server-sdk 6 (server); @ide/backoff peer dep.
- **AI:** openai SDK 6. Models: MODEL=gpt-5.4, MODEL_FAST=gpt-5.4-mini, MODEL_VISION=gpt-4o (vision turns only), whisper-1 (transcribe).
- **Supply-chain defense:** minimumReleaseAge: 1440 (1-day quarantine on every npm package).

4. Workflows (Replit-managed processes)

- **artifacts/api-server: API Server:** `pnpm --filter @workspace/api-server run dev` (esbuild bundle !' node start, mounted at /api).
- **artifacts/mobile: expo:** `pnpm --filter @workspace/mobile run dev` (expo start with Replit-specific env vars: EXPO_PACKAGER_PROXY_URL, EXPO_PUBLIC_DOMAIN, EXPO_PUBLIC_REPL_ID, EXPO_PUBLIC_CLERK_PUBLISHABLE_KEY, REACT_NATIVE_PACKAGER_HOSTNAME).
- **artifacts/mockup-sandbox: Component Preview Server:** `vite dev` (canvas iframes only).

5. Database Schema (Postgres via Drizzle, push not migrate)

Source: lib/db/src/schema/. Push command: pnpm --filter @workspace/db run push.

Table	Purpose	Key columns
users	One row per Clerk user	id (serial), clerk_user_id (unique), email, tier (free/pro/premium), expo_push_token, tz_offset_minutes, last_morning_nudge_date (YYYY-MM-DD), timestamps
user_state	Single per-user app blob	user_id (PK FK), goals (jsonb array), active_goal_id (text), account_prefs (jsonb), pending_draft (jsonb intake-in-progress), version (CAS counter), updated_at
conversations	Chat conversations	id, title, created_at
messages	Chat turns	id, conversation_id FK, role, content, created_at
subscriptions	Forward-looking billing record (RevenueCat / Stripe ready)	provider, product_id, status, current_period_end, store_transaction_id (unique with provider), raw jsonb
ai_usage	Per-call OpenAI ledger	route, model, input_tokens, output_tokens, latency_ms, status, error_message
analytics_events	Generic product/business events	event_type, payload jsonb

6. API Server (artifacts/api-server)

Entry: `src/index.ts` !' `src/app.ts` (62 lines). All routes mounted under `/api`.

6.1 Middlewares

- **clerkProxyMiddleware**: Passes Clerk session through dev proxy.
- **requireAuth**: Verifies Clerk token, upserts user row, sets `req.userId` (numeric DB id).
- **rateLimit**: `aiRateLimiter` applied at the `/atlas` router level.

6.2 Routes

`src/routes/health.ts`

- GET `/api/healthz`

`src/routes/me.ts` (218 lines) — single user-state surface

- GET `/api/me`: Public-safe identity + tier.
- GET `/api/me/state`: Returns full `MeStateResponse` (`goals[]`, `activeGoalId`, `accountPrefs`, `pendingDraft`, `version`).
- PUT `/api/me/state`: Optimistic concurrency via `expectedVersion`; returns 409 + latest on conflict.

`src/routes/atlas.ts` (1797 lines, the AI brain) — all under `/api/atlas/*`, requires auth + AI rate limiter

- POST `/generate-title`: Short brand-neutral goal title (NEW goals only).
- POST `/intake-questions`: Tailored intake form per goal type.
- POST `/intake-submit`: Synthesizes a `UserProfile` from answers.
- POST `/onboarding-chat`: Alternative chat-driven onboarding.
- POST `/roadmap`: Multi-phase Roadmap (`phases × milestones`, `strategy`, `riskAnalysis`, `totalWeeks`).
- POST `/daily-plan`: Today's `DailyTask[]` with `focusOfTheDay`, `coachNote`, `priority` (`critical/high/normal`).
- POST `/coach`: Free-form chat. Accepts `modelChoice` (`smart/fast`), optional `attachmentNote` (`filename` ack) OR `attachmentImage` { `base64Data`, `mimeType` } (real vision via `gpt-4o`, MIME allowlist, 5MB decoded cap). Returns `reply` + optional `actionSuggestion` (`Confirm/Cancel` cards) + optional `coachMemoryUpdate`.
- POST `/transcribe`: Multipart audio !' text via `Whisper`, 25 MB cap, `multiplex` memory storage.
- POST `/adapt`: Adaptive engine returning `easier/same/harder` recommendation.
- POST `/behavioral-profile`: Builds/evolves cumulative `BehavioralProfile` (`consistency`, `workload tolerance`, `motivation trend`, `focus style`, `peak hours`, `failure patterns`, `strengths`, `recommendedAdjustments`).
- POST `/evolve-roadmap`: Updates roadmap based on behavior + reflections; returns evolved roadmap + `phaseChanges[]` + `changeSummary` + `rationale`.
- POST `/push-token`: Registers Expo token + device tz offset on the user row.
- POST `/push-test`: One-shot self-test push.

6.3 Server-side libs

- `src/lib/logger.ts`: `pino` singleton (use `req.log` in routes).
- `src/lib/aiUsage.ts`: Wrapper that records every OpenAI call into `ai_usage`.
- `src/lib/pushScheduler.ts` (229 lines): In-process `setInterval` 60s tick. For each user with a token, computes local time from `tz_offset_minutes`; if local hour is in `[7,10)` and `last_morning_nudge_date` != today (local), picks the active goal, builds momentum-based body, sends via `expo-server-sdk`, stamps `last_morning_nudge_date` regardless of send result (prevents bad-token hourly retries).
- `src/app.ts`: JSON body limit raised 2 MB !' 8 MB (for base64 images), CORS, `cookie-parser`, mounts routes.

7. Mobile App (artifacts/mobile)

Entry: expo-router/entry. App config: app.json (newArch on, Clerk + push + audio plugins, expo-font plugin embeds .ttfs).

7.1 Routes (file-based, app/)

Auth group (auth)/

- **_layout.tsx**: Closed group (no escape to tabs).
- **sign-in.tsx**, **sign-up.tsx**, **verify.tsx**, **forgot-password.tsx**: Branded Clerk flows.

Top-level

- **_layout.tsx (262 lines)**: Root: ClerkProvider, QueryClientProvider, AtlasProvider, Inter + icon fonts via direct .ttf requires, AuthGate (auth + goal-gating + welcome-once-only logic), SafeAreaProvider, ErrorBoundary, status bar.
- **index.tsx**: Splash / route classifier.
- **welcome.tsx**: First-run welcome (shown exactly once per user).
- **intake.tsx**: Dynamic intake form rendering AI-generated questions.
- **generating.tsx**: Animated wait while roadmap+plan generate.
- **new-goal.tsx**: Add another goal (gated by tier).
- **replace-goal.tsx**: Swap an active goal at the tier limit.
- **plans.tsx**: Free/Pro/Premium tier picker (UI preview, no billing call).
- **+not-found.tsx**: 404.

Tab group (tabs)/

- **_layout.tsx**: Bottom tabs with Feather icons.
- **index.tsx**: Today (focus, daily tasks, momentum, smart/fast, voice, paperclip, chat composer).
- **roadmap.tsx**: Multi-phase roadmap with PhaseCards, evolve-roadmap entry point.
- **coach.tsx**: Full coach screen.
- **goals.tsx**: List/manage goals.
- **account.tsx**: Profile, plans link, sign out, behavioral profile refresh.

7.2 Providers

providers/AtlasProvider.tsx (1247 lines) — single source of truth for client app state:

- Hydrates from /api/me/state, falls back to AsyncStorage cache.
- Optimistic + coalesced PUT mutations with version CAS; on 409 re-syncs and retries.
- Tier refs: serverTierRef (authoritative, persisted) vs localTierOverrideRef (session-only Plans preview, never persisted).
- Exposes all goal/profile/roadmap/plan/coach/reflection mutations and selectors.
- Bootstraps push registration once per signed-in userId.

7.3 Components (components/)

AtlasButton, AtlasLogo, AwardToast, ChatBubble (with TTS speaker icon), ActiveGoalChip, AdaptiveEngineCard, EmptyState, ErrorBoundary, ErrorFallback, GoalCard, GoalListItem, GoogleGIcon, IntakeForm, KeyboardAwareScrollViewCompat, MomentumCard, PhaseCard, ProposedActionCard (Confirm/Cancel for AI task edits), ReflectionSheet, SectionHeader, SubscriptionCard, TaskCard, TaskDetailSheet (full-screen task view on tap).

7.4 Hooks (hooks/)

- **useColors**: Light/dark palette resolver.

- **useEvolveRoadmap**: Orchestrates /atlas/evolve-roadmap + state merge.
- **useTextToSpeech**: expo-speech (native) / window.speechSynthesis (web).
- **useVoiceRecorder**: expo-audio (native) / MediaRecorder (web), uploads to /atlas/transcribe.

7.5 Lib (lib/)

- **authErrors.ts**: Clerk error !' user-friendly copy.
- **awards.ts**: Streak/award logic.
- **momentum.ts**: Momentum band + body copy (mirrored server-side for push).
- **push.ts**: registerForPushAsync (permission, Expo token, tz offset, POST /atlas/push-token).
- **storage.ts**: Typed AsyncStorage cache snapshot.

7.6 Assets

- **assets/images/icon.png**: App icon, splash, adaptive icon.
- **assets/fonts/Feather.ttf, Ionicons.ttf, MaterialIcons.ttf**: Explicitly bundled via expo-font plugin (Android Expo Go fix).

8. Shared Libraries (lib/)

- **@workspace/api-spec:** openapi.yaml (single source of truth), Orval config; pnpm --filter @workspace/api-spec run codegen regenerates clients + zod.
- **@workspace/api-zod:** Generated Zod schemas (server-side input/output validation).
- **@workspace/api-client-react:** Generated React Query hooks (mobile data layer).
- **@workspace/db:** Drizzle schema, types, push CLI.
- **@workspace/integrations-openai-ai-server:** OpenAI server wrapper (chat, image, audio, batch) with p-limit + p-retry.
- **@workspace/integrations-openai-ai-react:** audio client helpers.

9. Auth & Routing Rules

Clerk session is the only signed-in concept. A custom AuthGate enforces:

- Signed-out !' only (auth) routes reachable.
- While AtlasProvider hydrating !' forced to /index.
- Signed-in with 0 goals !' trapped in welcome !' intake !' generating !' new-goal/replace-goal.
- Signed-in with goals.length > 0 landing on /welcome !' bumped to (tabs) (welcome is once-per-user).
- pendingDraft present !' resumes on /intake or /generating.

10. AI Coach Behavior

- Default model gpt-5.4; the Fast pill switches to gpt-5.4-mini; image attachments switch the single turn to gpt-4o.
- Coach proposes task edits as actionSuggestion !' renders ProposedActionCard with Confirm/Cancel; Confirm runs setActiveDailyPlan / updateActiveGoal !' schedulePush !' server PUT.
- Coach maintains coachMemory (summary + facts list capped at 20) updated via optional coachMemoryUpdate patches.
- AI auto-titles ONLY new goals (via /generate-title); never re-titles existing ones.
- Voice: mic record !' Whisper !' auto-send transcript; assistant bubbles get tap-to-speak; Voice on/off pill toggles auto-speak.

11. Push Notifications

- **Client:** registerForPushAsync after Clerk sign-in (web/simulator short-circuit), POSTs Expo token + UTC offset.
- **Server:** 60s in-process tick, local-morning window [7,10), dedupe via last_morning_nudge_date, momentum-aware body, stamp date even on send failure.
- **Self-test:** /atlas/push-test for manual verification.

12. Environment Variables / Secrets

- **DATABASE_URL**: Postgres connection.
- **CLERK_SECRET_KEY**, **CLERK_PUBLISHABLE_KEY**: Auth.
- **EXPO_PUBLIC_CLERK_PUBLISHABLE_KEY**: Mobile bundle-time.
- **AI_INTEGRATIONS_OPENAI_API_KEY**, **AI_INTEGRATIONS_OPENAI_BASE_URL**: OpenAI proxy.
- **SESSION_SECRET**: Express session signing.
- **PORT**, **REPLIT_DEV_DOMAIN**, **REPLIT_EXPO_DEV_DOMAIN**, **REPL_ID**: Replit-injected.

13. Build / Lint / Typecheck Commands

```
# Whole-repo typecheck (libs --build, leaves --noEmit)
pnpm run typecheck

# Regenerate API client + zod from openapi.yaml
pnpm --filter @workspace/api-spec run codegen

# Push DB schema to Postgres (no migrations)
pnpm --filter @workspace/db run push

# Mobile-only typecheck
pnpm --filter @workspace/mobile run typecheck

# Server bundle (esbuild !' dist/index.mjs)
pnpm --filter @workspace/api-server run build
```

14. Known Constraints / Conventions

- Never call service ports directly — always go through the shared proxy at localhost:80.
- Never use console.log in server code — use req.log or the singleton logger.
- All OpenAPI descriptions are single-line; nullable fields use oneOf: [type, "null"] (3.1 style).
- Mobile uses expo-router file-based routing, no manual nav stack.
- No architectural rewrites — UI mutations always flow through existing setActiveDailyPlan / updateActiveGoal !' server PUT path.
- Cloudflare CAPTCHA blocks Playwright e2e on auth flows — auth testing is manual on a real device.